

Prescribing Diagnostic Imaging

OMFS 553

Dr.Manila

Mission: To advance the science and art of healthcare through education, service, research, personal wellness, and social accountability.

Radiologic Examinations

- The design of the type and scope of the imaging examination should be guided by
 - The perceived nature or severity of an abnormality
 - The ability of the imaging technique to accurately reveal the characteristic diagnostic features of the abnormality
 - The amount of image detail required
 - The radiation dose to the patient



Guidelines for Ordering Imaging

- Form a framework for the image ordering practices of dentists
- Not hard and fast rules but rather principles upon which decisions may be made
- Includes appropriateness of the imaging examination to provide the needed diagnostic information and radiation-associated risks
- Apply guidelines only after a detailed review of the patient's medical and dental histories with a thorough clinical examination to identify signs or symptoms of possible disease
- Imaging findings should potentially help to direct the patient's diagnosis and/or treatment plan

Guidelines for Ordering Imaging

- Previous Diagnostic Images
 - helpful, regardless of when they were made
 - may be of value to demonstrate whether a condition (e.g., caries, periodontal disease) has remained unchanged, worsened, or shown resolution or healing
 - Images made relatively more recently may be adequate for the current diagnostic task, thus precluding the need to make images during the current visit
- Administrative Images
 - made for reasons other than diagnosis: Insurance company
 - do not benefit the patient, making such images should be avoided
 - NCRP 145: *“Administrative use of radiation to provide information not related to health of the patient shall not be permitted. Students shall not be permitted to perform radiographic exposures of patients, other students, or volunteers solely for purposes of their education or licensure.”*



Guidelines for Ordering Dental Radiographic Examinations

The following general categories of situations dictate a need for imaging

- Where there is clinical evidence of an abnormality that cannot be fully assessed by physical examination alone
 - hard swelling in the premolar region of the mandible with expansion of the buccal and lingual surfaces
- Where there is a high probability of disease that is not clinically evident
 - a patient may seek general dental care after not having seen a dentist for many years and clinical examination shows no evidence of caries

ADA Selection Criteria for Prescribing Dental Radiographs

Positive Historical Findings	Positive Clinical Signs or Symptoms
1. Previous periodontal or endodontic treatment	1. Clinical evidence of periodontal disease
2. History of pain or trauma	2. Large or deep restorations
3. Familial history of dental anomalies	3. Deep carious lesions
4. Post-operative evaluation of healing	4. Malposed or clinically impacted teeth
5. Remineralization monitoring	5. Swelling
6. Presence of implants or evaluation for implant placement	6. Evidence of dental/facial trauma
	7. Mobility of teeth
	8. Sinus tract ("fistula")
	9. Clinically suspected sinus pathology
	10. Growth abnormalities
	11. Oral involvement in known or suspected systemic disease
	12. Positive neurologic findings in the head and neck
	13. Evidence of foreign objects
	14. Pain and/or dysfunction of the temporomandibular joint
	15. Facial asymmetry
	16. Abutment teeth for fixed or removable partial prosthesis
	17. Unexplained bleeding
	18. Unexplained sensitivity of teeth
	19. Unusual eruption, spacing or migration of teeth
	20. Unusual tooth morphology, calcification or color
	21. Unexplained absence of teeth
	22. Clinical tooth erosion
	23. Peri-implantitis



ADA Guidelines for Prescribing Dental Radiographs

PATIENT AGE AND DENTAL DEVELOPMENTAL STAGE		
Type of Encounter	Child With	
	Child With Primary Dentition (Before Eruption of First Permanent Tooth)	Transitional Dentition (After Eruption of First Permanent Tooth)
New patient ^a being evaluated for oral diseases	Individualized radiographic exam consisting of selected periapical/occlusal views and/or posterior bitewings if proximal surfaces cannot be visualized or probed. Patients without evidence of disease and with open proximal contacts may not require a radiographic examination at this time	Individualized radiographic exam consisting of posterior bitewings with panoramic exam or posterior bitewings and selected periapical images
Recall patient ^a with clinical caries or at increased risk for caries ^b	Posterior bitewing exam at 6- to 12-month intervals if proximal surfaces cannot be examined visually or with a probe	
Recall patient ^a with no clinical caries and not at increased risk of developing caries ^b	Posterior bitewing examination at 12- to 24-month intervals if proximal surfaces cannot be examined visually or with a probe	
Recall patient ^a with periodontal disease	Clinical judgment as to the need for and type of radiographic images for the evaluation of periodontal disease. Imaging may consist of but is not limited to selected bitewing and/or periapical images of areas in which periodontal disease (other than nonspecific gingivitis) can be demonstrated clinically	
Patient (new and recall) for monitoring of dentofacial growth and development and/or assessment of dental/skeletal relationships	Clinical judgment as to need for and type of radiographic images for evaluation and/or monitoring of dentofacial growth and development or assessment of dental and skeletal relationships	
Patient with other circumstances, including but not limited to proposed or existing implants, other dental and craniofacial pathosis, restorative/endodontic needs, treated periodontal disease, and caries remineralization	Clinical judgment as to need for and type of radiographic images for evaluation and/or monitoring of these conditions	



ADA Guidelines for Prescribing Dental Radiographs-Contd

PATIENT AGE AND DENTAL DEVELOPMENTAL STAGE		
Adolescent With Permanent Dentition (Before Eruption of Third Molars)	Adult, Dentate or Partially Edentulous	Adult, Edentulous
Individualized radiographic exam consisting of posterior bitewings with panoramic exam or posterior bitewings and selected periapical images; full-mouth intraoral radiographic exam is preferred when patient has clinical evidence of generalized dental disease or a history of extensive dental treatment		Individualized radiographic exam based on clinical signs and symptoms
Posterior bitewing exam at 6- to 12-month intervals if proximal surfaces cannot be examined visually or with a probe	Posterior bitewing examination at 6- to 18-month intervals	Not applicable
Posterior bitewing exam at 18- to 36-month intervals	Posterior bitewing exam at 24- to 36-month intervals	Not applicable
Clinical judgment as to the need for and type of radiographic images for the evaluation of periodontal disease. Imaging may consist of, but is not limited to, selected bitewing and/or periapical images of areas in which periodontal disease (other than nonspecific gingivitis) can be demonstrated clinically		Not applicable
Clinical judgment as to need for and type of radiographic images for evaluation and/or monitoring of dentofacial growth and development or assessment of dental and skeletal relationships. Panoramic or periapical exam to assess developing third molars	Usually not indicated for monitoring of growth and development. Clinical judgment as to the need for and type of radiographic images for evaluation of dental and skeletal relationships	
Clinical judgment as to need for and type of radiographic images for evaluation and/or monitoring of these conditions		



AAOMR-AAE recommendations for the use of CBCT

1. **Intraoral radiographs** should be considered the imaging **modality of choice in the evaluation** of the endodontic patient
2. **Limited field-of-view** (FOV) cone beam computed tomography (CBCT) should be considered the imaging modality of choice for diagnosis in patients who present with **contradictory or nonspecific clinical signs and symptoms** associated with **untreated or previously endodontically treated teeth**
3. **Limited FOV CBCT** should be considered the imaging modality of choice for initial treatment of teeth with the **potential for extra canals** and **suspected complex morphology**, such as mandibular anterior teeth, maxillary and mandibular premolars and molars, and dental anomalies.
4. If a **preoperative CBCT has not been taken**, limited FOV CBCT should be considered as the imaging modality of choice for intra-appointment identification and **localization of calcified canals**.
5. **Intraoral radiographs** should be considered the imaging modality of choice for **immediate postoperative imaging**.
6. **Limited FOV CBCT** should be considered the imaging modality of choice if clinical examination and two-dimensional intraoral radiography are inconclusive in the **detection of vertical root fracture** (VRF)



AAOMR-AAE recommendations for the use of CBCT

7. **Limited FOV CBCT** should be the imaging modality of choice in evaluating the **nonhealing of previous** endodontic treatment to help determine the need for further treatment, such as **nonsurgical or surgical treatment or extraction**.
8. **Limited FOV CBCT** should be the imaging modality of choice for **nonsurgical retreatment** to assess endodontic treatment complications, **such as overextended root canal obturation material, separated endodontic instruments, and localization of perforations**.
9. **Limited FOV CBCT** should be considered as the imaging modality of choice for **presurgical treatment planning to localize root apex/apices and to evaluate the proximity to adjacent anatomic structures**.
10. **Limited FOV CBCT** should be considered as the imaging modality of choice **for surgical placement of implants**.
11. **Limited FOV CBCT** should be considered the imaging modality of choice for diagnosis and management of limited dentoalveolar trauma, **root fractures, luxation, and/or displacement of teeth and localized alveolar fractures in the absence of other maxillofacial or soft tissue injury** that may require other advanced imaging modalities.
12. **Limited FOV CBCT** is the imaging modality of choice in the localization and differentiation of **external and internal resorptive defects** and the determination of appropriate treatment and prognosis.



AAOMR recommendations for the use of CBCT in Orthodontics

1. To assess **anomalies** in the structure and **eruption** of teeth
2. To evaluate **dental-alveolar boundaries**
3. To assess **craniofacial skeletal asymmetry**
4. To assess **maxillomandibular discrepancies**
5. To evaluate the **temporomandibular joints**
6. To assess **airway morphology**
7. To **plan for** combined orthodontic and orthognathic **surgical management**
8. To identify optimal **locations** for **temporary anchorage devices**
9. To **assess skeletal and dental changes** produced by maxillary expanders



Application of CBCT in Dentistry

Justification and referral criteria CBCT

Justification and referral criteria

- All CBCT examinations must be justified on an individual basis by demonstrating that the potential benefits to the patients outweigh the potential risks. **CBCT examinations should potentially add new information to aid the patient's management.** A record of the Justification process must be maintained for each patient
- CBCT should not be selected unless a history and clinical examination have been performed. **"Routine" or "screening" imaging is unacceptable practice**
- When referring a patient for a CBCT examination, the referring dentist must **supply sufficient clinical information** (patient history and results of examination) to allow the CBCT Practitioner to perform the Justification process

- Endodontic applications of CBCT
 - Differentiation of pathosis from normal anatomy
 - Relationships with important anatomical structures
 - Aiding management of dens invaginatus and aberrant pulpal anatomy
 - External resorption
 - Internal resorption
 - Lateral root perforation by a post
 - Accessory canal identification
 - Surgical management of fractured instrument
 - Aiding surgical endodontic planning

Justification and referral criteria

- **Orthodontic applications of CBCT**
 - Cleft palate assessment
 - Tooth position and localization
 - Resorption related to impacted teeth
 - Measuring bone dimensions for mini-implant placement
 - For rapid maxillary expansion
 - 3D cephalometry
 - Surface imaging integration
 - Age assessment
 - Investigation of orthodontic-associated paresthesia

Justification and referral criteria

- **CBCT in dental trauma**
 - Root fractures
 - Luxation injuries
 - Avulsion
 - Root resorption as a post-trauma complication
- CBCT may be indicated for the localized assessment of an impacted tooth (including consideration of resorption of an adjacent tooth) where the current imaging method of choice is conventional dental radiography and when the information cannot be obtained adequately by lower dose conventional (traditional) radiography

Justification and referral criteria

Assessment of periapical disease

- CBCT is not indicated as a standard method for identification of periapical pathosis
- Limited volume, high resolution CBCT may be indicated for periapical assessment, in selected cases, when conventional radiographs give a negative finding when there are contradictory positive clinical signs and symptoms
- Where CBCT images include the teeth, care should be taken to check for periapical disease when performing a clinical evaluation (report)
 - Properly validated in vivo studies impossible
 - Lab and animal studies possible and suggest improved sensitivity of CBCT
 - Simple comparative clinical studies
 - “34% of lesions were missed by periapical radiography” (Low et al 2008)



Justification and referral criteria

- **Surgical applications**
 - Where conventional radiographs suggest a direct inter-relationship between a mandibular third molar and the mandibular canal, and when a decision to perform surgical removal has been made, CBCT may be indicated
 - CBCT may be indicated for pre-surgical assessment of an unerupted tooth in selected cases where conventional radiographs fail to provide the information required
- **Bony pathosis**
 - Where it is likely that evaluation of soft tissues will be required as part of the patient's radiological assessment, the appropriate initial imaging should be MSCT or MR, rather than CBCT
 - Limited volume, high resolution, CBCT may be indicated for evaluation of bony invasion of the jaws by oral carcinoma when the initial imaging modality used for diagnosis and staging (MR or MSCT) does not provide satisfactory information

Justification and referral criteria

- **Surgical applications: Facial trauma**
 - For maxillofacial fracture assessment, where cross-sectional imaging is judged to be necessary, CBCT may be indicated as an alternative imaging modality to MSCT where radiation dose is shown to be lower and soft tissue detail is not required
- **Surgical applications: Orthognathic surgery**
 - CBCT is indicated where bone information is required, in orthognathic surgery planning, for obtaining three-dimensional datasets of the craniofacial skeleton
- **Surgical applications: Temporomandibular joint**
 - Where the existing imaging modality for examination of the TMJ is MSCT, CBCT is indicated as an alternative where radiation dose is shown to be lower

Justification and referral criteria

- Periodontal applications of CBCT
 - CBCT is not indicated as a routine method of imaging periodontal bone support
 - Limited volume, high resolution CBCT may be indicated in selected cases of infra-bony defects and furcation lesions, where clinical and conventional radiographic examinations do not provide the information needed for management
 - Where CBCT images include the teeth, care should be taken to check for periodontal bone levels when performing a clinical evaluation (report)

Justification and referral criteria

- Implant dentistry

Special indications for cross-sectional imaging

Maxilla	Single tooth	a. incisive canal b. descent of maxillary sinus c. clinical doubt about shape of alveolar ridge
	Partially dentate	a. descent of maxillary sinus b. clinical doubt about shape of alveolar ridge
	Edentulous	a. descent of maxillary sinus b. clinical doubt about shape of alveolar ridge
Mandible	Single tooth	a. clinical doubt about position of mandibular canal b. clinical doubt about shape of alveolar ridge
	Partially dentate	a. clinical doubt about position of mandibular canal or mental foramen b. clinical doubt about shape of alveolar ridge
	Edentulous	a. severe resorption b. clinical doubt about shape of alveolar ridge c. clinical doubt about position of mandibular canal if posterior implants are to be placed

Justification and referral criteria

- **Implant dentistry**
 - CBCT is indicated for cross-sectional imaging prior to implant placement as an alternative to existing cross-sectional techniques where the radiation dose of CBCT is shown to be lower
 - For cross-sectional imaging prior to implant placement, the advantage of CBCT with adjustable fields of view, compared with MSCT, becomes greater where the region of interest is a localized part of the jaws, as a similar sized field of view can be used



Final points

“CBCT should only be used when the question for which imaging is required cannot be answered adequately by lower dose conventional radiography”

Is it likely to make any difference to my treatment plan?



Imaging considerations in the Absence of a Positive Finding

- The prevalence of an abnormality in the jaws
- The probability that such an abnormality would not produce a clinically detectable sign or symptom
- The ability of the imaging modality to detect the disease
- Whether the detection of such an abnormality would influence the management of the patient

Special Considerations

Pregnancy

- When radiation protection practices are followed including collimation, fast or digital receptors, and protective shielding, the doses from diagnostic dentomaxillofacial radiology are negligible
- Fetal dose is 42000-fold lower than the threshold dose for deterministic effects on the embryo and fetus
- radiologic imaging in all patients is predicated on there being a diagnostic need, the guidelines apply equally to pregnant patients as well as to those who are not pregnant
- It is, however, recommended to use a protective thyroid collar when it does not obstruct an essential anatomic region of the image

Special Considerations

Radiation Therapy

- Patients may receive 50 Gy or more during treatment
- Radiation doses from dental imaging procedures are relatively insignificant
- Patients must be carefully monitored because they are at special risk for dental diseases